

Heterogeneous Integration of Low Loss Si₃N₄ Waveguides and High-Speed Si-Ge Photodiodes

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Abstract We demonstrate a Ge photodiode heterogeneously integrated on a low-loss Si₃N₄ 200mm platform, achieving a 3dB bandwidth beyond 50GHz at 1550nm. This approach enables high-speed photodetection while leveraging low-loss waveguides, compatible with Kerr frequency comb generation, paving the way for highly scalable DWDM transceivers. ©2026 The Author(s)

Introduction

High-speed photodetectors are key components for next-generation optical interconnects, microwave photonics and coherent communication systems, where increasing data rates impose stringent requirements on bandwidth and integration density [1],[2]. Transceivers based on frequency combs are attractive from that standpoint, with ultra-dense optical data transfer feasible on a single chip with a single pump laser source [3]. However, frequency combs generally require very low propagation losses waveguides, achievable in stand-alone Si₃N₄ platforms [4].

Germanium-on-silicon (Ge-on-Si) photodiodes, on the other hand, showed excellent performance in silicon-based platforms, offering high responsivity and large bandwidth at telecommunication wavelengths around 1550 nm [5],[6]. Several works reported high-speed photodiodes exceeding tens of gigahertz, with bandwidths beyond 60 GHz [7],[8]. However, these demonstrations were primarily realized on silicon-on-insulator platforms, with high propagation losses. In contrast, silicon nitride (SiN) platforms have recently attracted strong interest due to their ultra-low propagation losses and suitability for large-scale photonic circuits and microwave photonic systems [9],[10]. Despite these advantages, the integration of high-speed photodetectors on very low-loss SiN remains relatively unexplored, mainly due to integration challenges: indeed, low loss SiN waveguides require a high temperature Low Pressure-Chemical Vapour Deposition (LPCVD) and even higher annealing temperatures which are prohibited in the presence of Ge epitaxial layers (melting temperature of Ge: 938°C).

We demonstrate here the integration of a vertical silicon-germanium PIN photodiode on a low-loss silicon nitride platform based on the direct bonding of a Silicon-On-Insulator (SOI) wafer onto the patterned Si₃N₄ waveguides. The proposed device leverages a vertical architecture to ensure efficient optical absorption while

preserving high-speed operation, with a measured 3 dB bandwidth beyond 50 GHz.

Fabrication and design

The fabrication is performed on the 200mm pilot line of CEA-Leti. The first steps consist in the fabrication of 800nm-thick LPCVD Si₃N₄ waveguides [10]. Waveguides are then annealed at 1200°C and planarized. Those Si₃N₄ waveguides exhibit losses as low as 0.08dB/cm at 1550nm and features frequency comb generation in ring resonators. SOI wafers with 220nm thick Si overlayers are then reported by direct bonding. After donor substrate removal, Si waveguides are doped, patterned and encapsulated. Ge is grown on top of Si by selective epitaxy in an oxide cavity [11], planarized and oxide encapsulated. SEM cross sections are shown in Fig. 1. The bonding interface at the bottom of the Si layer is no longer visible. Finally, an AlCu backend-of-line and W vias are used for electrical connection.

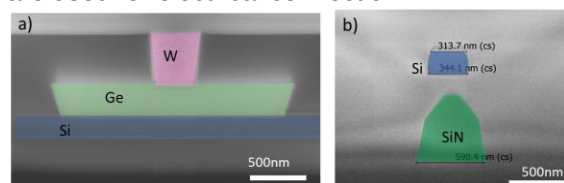


Fig. 1: Cross-sectional Scanning Electron Microscopy (SEM) images of a) the Ge on Si photodiode and b) the Si₃N₄ -Si transition.

The optical signal is transferred first from the Si₃N₄ waveguide to the Si waveguide through an adiabatic taper. The insertion loss measurement map of this transition is shown in Fig. 2 (0.26dB median loss). Light is then absorbed in the Ge region through a butt coupled configuration.

This approach is particularly attractive for integrated photonic circuits, as the low-loss SiN platform yields efficient distribution of optical power over long routing segments, enabling highly parallel photodiodes integration.

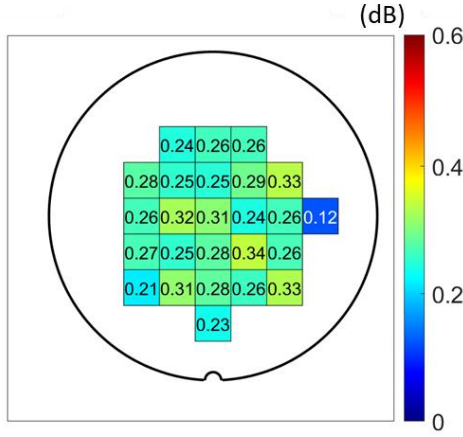


Fig. 2: Wafer map measurement of the SiN-Si transition insertion losses at 1550nm

We measured around 8dB loss for one grating coupler (GC) at 1550nm from a dedicated test structure. The optical signal generated by an external laser source (EXFO) was coupled to the device thanks to a surface GC. We swept the input optical power at the photodiode input (de-embedded with the GC loss and the SiN-Si transition loss) and extracted the photodiode responsivity from the linear fit between the measured photocurrent and the input optical power.

As shown in Fig. 3, the photodiode architecture is based on a vertical PIN structure. The device consists of an intrinsic germanium absorption region sandwiched between a p-doped Ge layer and an n-doped silicon layer, enabling efficient carrier separation under reverse bias. This vertical configuration presents a major advantage: the electron transit time is given by the Ge thickness which is much easier to control than a Ge width in a lateral configuration, potentially resulting in much higher bandwidths. Accordingly, the device geometry was carefully designed by controlling the thickness of each layer. The thicknesses of the p-Ge (t_p), i-Ge (t_i), and n-Si (t_n) regions were 50, 300, and 220 nm, respectively.

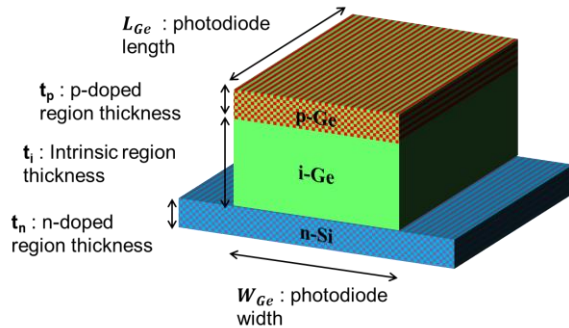


Fig. 3 Schematics of the vertical Ge-on-Si PIN photodiode.

To investigate the trade-off between responsivity and bandwidth, the photodiode width (W_{Ge}) was fixed to 2 μm while the device length (L_{Ge}) was varied. Several photodiodes were fabricated

with 10, 15, 25, and 50 μm lengths. Shorter devices should exhibit reduced capacitance, enabling higher bandwidth, while longer devices should yield increased optical absorption and responsivity.

Static and Radio-Frequency measurements

Static characteristics were first analyzed to assess the dark current and responsivity behavior of the fabricated photodiodes. All measurements were performed at room temperature. Fig. 4 shows the measured dark current I_{dark} as a function of the applied bias voltage. The shorter device exhibits a dark current of 480 nA at -1 V, which is relatively high compared to typical values reported for Ge-on-Si photodiodes [8]. To further investigate the origin of this elevated dark current, the dependence of I_{dark} on photodiode geometry was analysed.

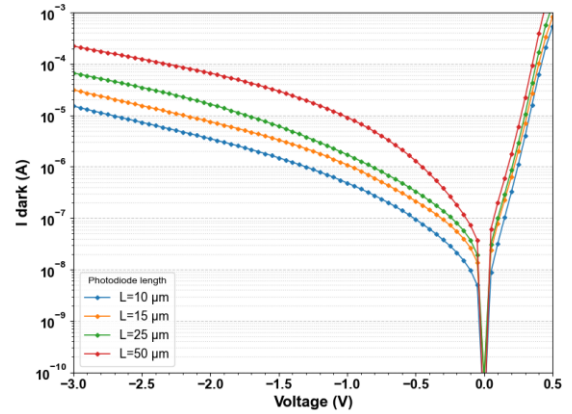


Fig. 4 Dark current I-V characteristics of the Ge-on-Si photodiode.

The dark current dependency with the photodiode area ($W_{Ge} \times L_{Ge}$) was extracted to identify the major noise contributor. We obtained a quasi-linear relationship suggesting that bulk generation dominated over surface generation [12]. The bulk Shockley-Read-Hall (SRH) leakage is due to the presence of threading dislocations in the Ge layer (a few 10^7 cm^{-2}) caused by the 4.2% Ge/Si lattice mismatch [13]. Under high reverse bias, an increase in the dark current is observed as energy bands bend, enabling carriers to transit from the valence band to the conduction band. The presence of additional crystalline defects may also enhance the dark current due to trap-assisted tunneling (TAT).

Fig. 5 shows the measured responsivity of the photodiodes at 1550 nm. The obtained responsivity (around 0.55A/W) remains lower than typical values for Ge-on-Si photodiodes [8]. This behaviour may be attributed to a combination of limited optical absorption, coupling efficiency, and material quality, which will be further investigated.

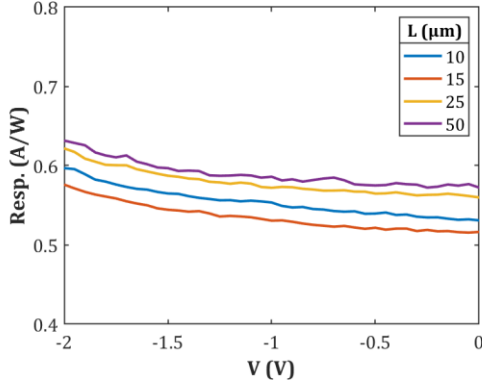


Fig. 5 Responsivity as a function of reverse bias

Following static and optoelectronic (OE) characterizations, high-speed measurements were performed using a classical RF experimental set-up. Light was provided by an external continuous-wave laser operating at 1550 nm and coupled into the device, with a variable optical attenuator placed before the grating couplers to control the injected optical power. The photodiodes were biased using GSG probes (MPI Titan) with a 100 μm pitch. The generated photocurrent was measured using a 110 GHz Lightwave Component Analyzer (Keysight N4372E), providing the frequency response through S21 measurements. A de-embedding procedure based on open and short reference structures was applied to remove parasitic pad and interconnect contributions [14].

Fig. 6 shows the normalized frequency response (S21) of the photodiodes for different device lengths under a reverse bias of -3 V . The optical input power is 0.5 mW at the fiber output. The measured 3 dB OE bandwidth as a function of photodiode length is shown in Fig. 7 for different reverse bias voltages.

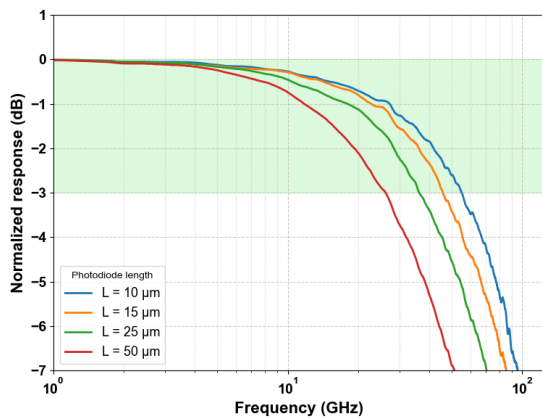


Fig. 6 Normalized S21 frequency response of the Ge-on-Si photodiodes for different device lengths measured at 1550 nm, with an input optical power of 0.5 mW and a reverse bias of -3 V . The shaded region highlights the -3 dB bandwidth.

The shortest device (10 μm) achieves the highest bandwidth with 55 GHz at -3 V ; a clear bandwidth reduction is observed as the device length increases. Furthermore, the OE bandwidth

increases with voltage because of the decrease in capacitance with the junction being depleted. The longest device (50 μm) exhibits the lowest bandwidth, which is attributed to increased capacitance. These results further confirm the trade-off between bandwidth and optical absorption in the proposed vertical Ge-on-Si photodiode architecture.

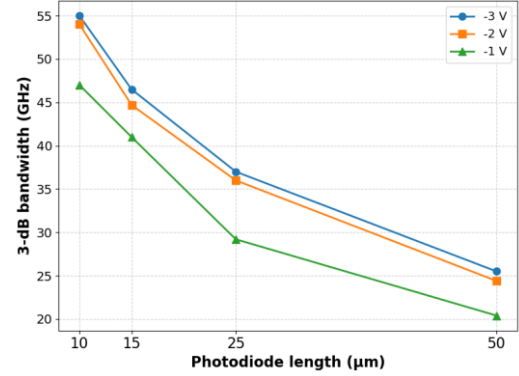


Fig. 7 Measured 3-dB bandwidth as a function of photodiode length for different reverse bias voltages, with an input optical power of 0.5 mW.

Conclusions

In this work, we demonstrated the integration of vertical silicon-on-germanium (Ge-on-Si) PIN photodiodes on low-loss silicon nitride platforms using a 200 mm wafer process flow. The proposed architecture, based on direct bonding, enabled the co-integration of high speed photodetection with low losses Si_3N_4 waveguides compatible with frequency comb generation.

A comprehensive static and dynamic characterization were performed at 1550 nm wavelength. Static measurements showed relatively high dark current and limited responsivity, likely due to defect-related generation in the germanium layer. Further optimization of the material quality and optical coupling is therefore required. Despite these limitations, devices exhibited excellent high-frequency performance. A maximum 3-dB optoelectronic bandwidth exceeding 50 GHz was achieved for a 10 μm -long photodiode under a reverse bias of -3 V .

These results demonstrate the strong potential of combining low-loss Si_3N_4 photonic platforms with high-speed Ge-on-Si photodetectors, paving the way for scalable integrated optical transceivers based on Kerr frequency combs and advanced microwave photonic systems.

Acknowledgements

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